Estimate end of queue position

Queueing is expected for 'through' methods at stop locations where PTCDs or traffic controllers are positioned, sometimes resulting in collision. Collision can occur when the stationary queue extends past the PREPARE TO STOP sign location, most commonly when speed is greater than 70 km/h or the sight distance of approaching traffic to the end of the queue is:

- less than two times the speed limit in open road areas
- less than 1.5 times the speed limit in built-up areas.

To estimate queue length:

- Count the number of average and oversized vehicles that pass the PTCD/traffic controller position for five (5) minutes.
- Consider whether the majority of vehicles have been average or oversized (i.e. trucks). This will influence the 'multiplier' column used in Table 4.3.
- Multiply the number of vehicles counted by the number in the chosen 'multiplier' column (Ma for mostly average sized vehicles, or Mo for mostly oversized vehicles) using the maximum stop time required at the specific worksite.
- If you are unsure of the maximum required stop time or whether to use the 'average' or 'oversized' multiplier, seek assistance from a competent person or road authority.
- Use the formula below to calculate the estimated queue length:

  (number of average vehicles × Ma) + (number of oversized vehicles × Mo) = queue length

If more accurate data is available (e.g. traffic counts), this should be used instead of counting vehicles for five (5) minutes.

An estimated end-of-queue position is to be determined for the approach to each traffic control station and is to be based on the maximum expected traffic flow on that approach during the time traffic control will be in operation.

The count or estimate of the number of average and heavy vehicles during a five-minute period at a site may be completed using the following in order of preference:

1. Actual five-minute count of vehicles during the peak time the site will be occupied. This five-minute count is based on the vehicles approaching the selected traffic control station from the approach to be controlled by that station (not a sum of both directions of traffic). Consideration of peak traffic flow direction may be needed.
2. If a five-minute count is not possible, use annual average daily traffic (AADT) values with hourly breakdowns and percentage heavy vehicle data. To estimate the five-minute count, select the peak hourly period during the time the site will be occupied and divide by 12 to get an estimated five-minute value. Divide this by two if the AADT is for a two-way road. Use the percentage heavy vehicles information with this value to estimate the number of heavy vehicles for this five-minute period.